13.2 Counting Voles

WORKSHEET

The question

STEP II 2000, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The population has N voles, 200 of them marked. A random sample of 200 is taken. What is the probability that exactly 11 are marked?

STAGE 1 The likelihood

Covers: the probability p_N

Ask yourself Choose 11 of the marked voles and 189 of the unmarked ones, out of all equally likely samples of 200. Then write out the binomial coefficients that involve N as factorials.

STAGE 2 Maximising over a whole number

Covers: maximising p_N

Pause and try N is a whole number, so compare p_{N+1} with p_N (or p_N with p_{N-1}). Most of the factorials cancel. When is the ratio bigger than 1?

STAGE 3 A third sample

Covers: the third sample

Ask yourself After the second sample, how many voles are marked? With your estimate of N, how many are unmarked? Write the probability that a third sample of 200 contains exactly j marked voles. What do these probabilities add up to?

What's really going on

Maximum likelihood

The principle of *maximum likelihood*, developed by Ronald Fisher in the 1910s and 1920s, says: estimate an unknown parameter by the value that makes the observed data most probable. It is by far the most widely used method of estimation, from fitting curves to data to training the neural networks of machine learning, whose "loss functions" are usually negative log-likelihoods. Fisher showed that, for large samples, maximum likelihood estimates are approximately normal and as precise as any reasonable estimator can be, provided the model is well behaved in ways that hold for most everyday models. There are important exceptions. In the next section (Section 13.3) the unknown is the largest possible value, an endpoint of the distribution, and there the maximum likelihood estimate behaves quite differently: its error is not approximately normal, and it shrinks like $\frac{1}{n}$ rather than $\frac{1}{\sqrt{n}}$.

How precise is the estimate?

Figure 13.2.1 (in the full section online) shows that the likelihood is broad: 11 recaptures is not much information. A rough calculation: the number of marked voles in the second sample is approximately Poisson with mean $\frac{200 \times 200}{N}$, so its standard deviation is about $\sqrt{11} \approx 3.3$, a relative error of about 30%, which carries over to the estimate of N. To estimate N to within 10%, the biologists would need about 100 recaptures. Statisticians also correct for the small bias of the simple estimate, using the *Chapman estimator* $\frac{(M+1)(C+1)}{R+1} - 1$, where M and C are the sizes of the two catches and R the number recaptured.

Laplace counts France

Laplace used the same idea in 1802 to estimate the population of France without a census. He knew the number of births registered in the whole country in a year, and, for a sample of communes, both the population and the births. Assuming the ratio of population to births was the same everywhere, he estimated the population of France at about 28 million, and even gave a measure of the uncertainty. The method was rediscovered for animal populations by C. G. J. Petersen, who marked plaice in Danish fjords in the 1890s, and by Frederick Lincoln, who used bird rings in the 1930s; it is still often called the *Lincoln–Petersen* estimator.

Try this next

1. **General formula.** If M animals are marked and a second sample of C contains R marked, show that the maximum likelihood estimate of N is the integer part of $\frac{MC}{R}$.

2. **Vandermonde.** Prove $\sum_{j} \binom{a}{j}\binom{b}{c-j} = \binom{a+b}{c}$ by counting committees of c people chosen from a women and b men.

3. **Coin tosses.** In n tosses of a coin with unknown probability p of heads, h heads are observed. Show that the maximum likelihood estimate of p is $\frac{h}{n}$.

4. **No recaptures.** What happens to the maximum likelihood estimate if the second sample contains no marked voles? What does that tell you about the method?

Further reading

- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on estimation and uncertainty.
- **George Seber, *The Estimation of Animal Abundance*** (Blackburn). The standard reference on capture–recapture.
- **John Rice, *Mathematical Statistics and Data Analysis*** (Duxbury). A first course in mathematical statistics, including maximum likelihood.

The full guided solution is in Section 13.2:
https://beyondstep.keithmcnulty.org/ch13/voles.html